

SM-A520F_DS_REV0.4A

2016.11.09

sheet01 : GSM / WCDMA / LTE RF

sheet02 : TRANCEIVER, DRX

sheet03 : JOON(EXYNOS7880), eMMC

sheet04 : SENSORS, GPS, BT/WIFI, NFC, BTP, FM RADIO

sheet05 : PMIC(Main,IF), MST, KEY, MOTOR

sheet06 : Audio, SD/SIM, IF Conn'

sheet07 : CAMERA, DISPLAY, SHEILD CAN, TDMB

Project	
Product	
Model	
Doc. Title	
Doc. No.	
Rev.	
Issue	
Page	

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO.'S PROPERTY.
DO NOT DISCLOSE TO ANY UNAUTHORIZED PERSONS
EXCEPT AS AUTHORIZED BY SAMSUNG.

RF PART 2

HIGH BAND PA

FDD LTE: B7
TDD LTE: B38 / 40 / 41

MMMB PA

2G: EGSM / DCS / PCS / GSM850
3G: B1 / 2 / 4 / 5 / 8
FDD LTE: B1 / 2 / 3 / 4 / 5 / 8 / 17 / 20 / 28

SUB ANT2 (LB)

SUB ANT1 (MB/HB)

SUB ANT2 XGND

SUB ANT2 XGND

B38/41

B2

B40

B28

B29

TRANSCEIVER

26MHz TCXO

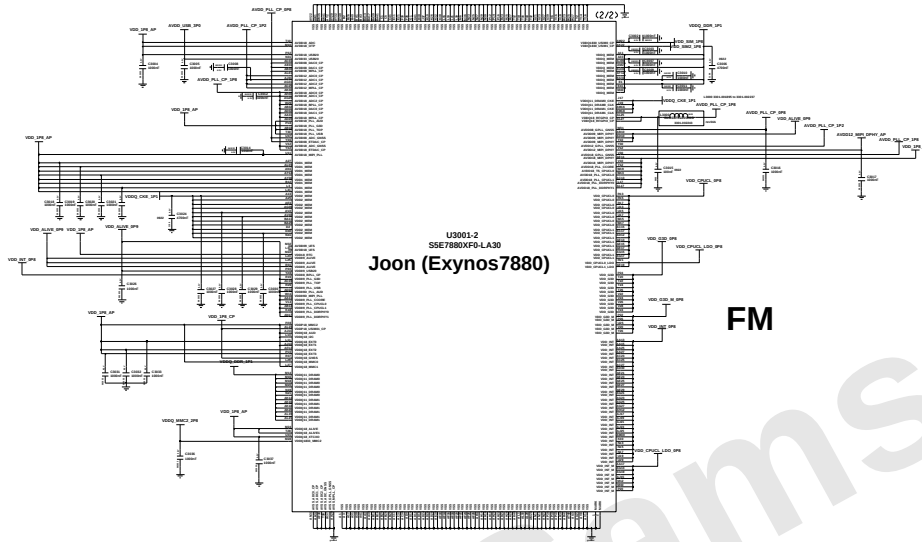
FOR CANADA

Engineer	RF
Drawn by	
Check by	
Doc. Ctrl. by	
Mfg. Engr. by	
REV	Drawing Number
Sheet	

EXYNOS7880(JOON)

POP LP4 3G : 1st Micron
2nd SEC

Salt Water detect
MUX

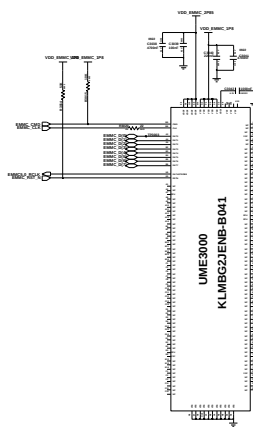


FM

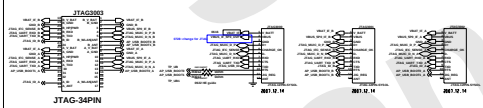
Joon (Exynos7880)

eMMC 32GB 5.1

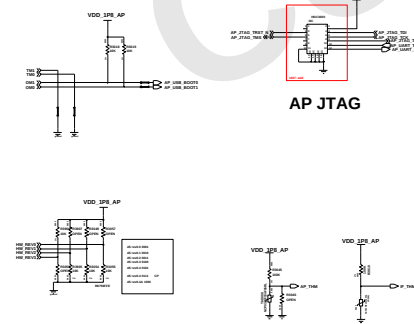
1st : SEC
2nd : TOSHIBA



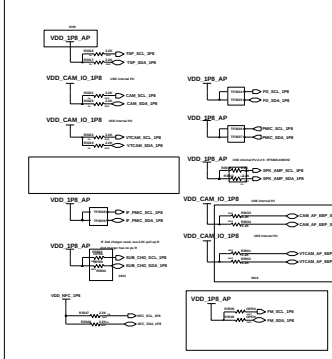
PCB JTAG



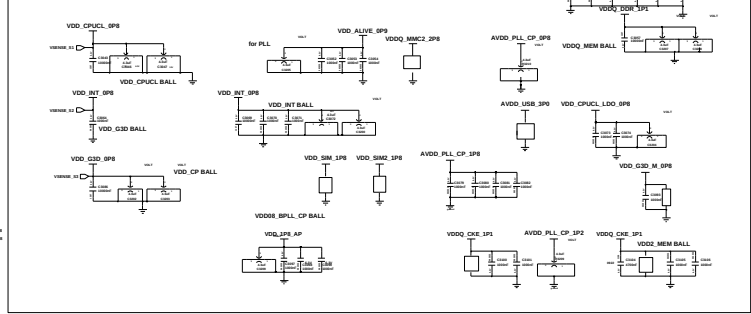
AP JTAG



I2C Pull-up Register



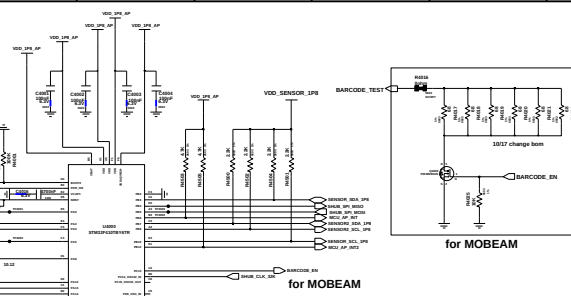
AP Decap



SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL INFORMATION. IT IS TO BE KEPT UNDER STRICT CONTROL AND NOT DISCLOSED TO ANY OTHER PERSONS EXCEPT AS AUTHORIZED BY SAMSUNG.

SENSORS

Debug port



SENSOR HUB
(2017 A)

Hall IC

near to IF con AREA
Flip open/close

View Cover Detect

6/27

Sensor LDO

PROXY+RGB FPCB
TMD3725(AMS)

Magnetic Sensor

7/26 change

Pressure Sensor

6 Axis Sensor(ACC + GYRO)

BTP + HOME KEY

NC BTP DC-DC

BTP OCP and PWR

BTP IF Conn

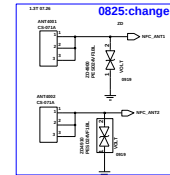
BT/WIFI

REV0.1
0823: change

GPS

NFC

FM



REVISION	DESCRIPTION	DATE
001	SENSOR, WIFI/BT/GPS/NFC	2016.11.14
002	0825: change	2016.11.14

Main_PMIC

**U5807
SZMPU05X01-6230**

Buck1 & Buck2 inductor -> Induct max >= 2.5A , Induct max >= 2.5A

Pin List:

Pin	Function
1	VDD
2	VSS
3	VDD
4	VSS
5	VDD
6	VSS
7	VDD
8	VSS
9	VDD
10	VSS
11	VDD
12	VSS
13	VDD
14	VSS
15	VDD
16	VSS
17	VDD
18	VSS
19	VDD
20	VSS
21	VDD
22	VSS
23	VDD
24	VSS
25	VDD
26	VSS
27	VDD
28	VSS
29	VDD
30	VSS
31	VDD
32	VSS
33	VDD
34	VSS
35	VDD
36	VSS
37	VDD
38	VSS
39	VDD
40	VSS
41	VDD
42	VSS
43	VDD
44	VSS
45	VDD
46	VSS
47	VDD
48	VSS
49	VDD
50	VSS
51	VDD
52	VSS
53	VDD
54	VSS
55	VDD
56	VSS
57	VDD
58	VSS
59	VDD
60	VSS
61	VDD
62	VSS
63	VDD
64	VSS
65	VDD
66	VSS
67	VDD
68	VSS
69	VDD
70	VSS
71	VDD
72	VSS
73	VDD
74	VSS
75	VDD
76	VSS
77	VDD
78	VSS
79	VDD
80	VSS
81	VDD
82	VSS
83	VDD
84	VSS
85	VDD
86	VSS
87	VDD
88	VSS
89	VDD
90	VSS
91	VDD
92	VSS
93	VDD
94	VSS
95	VDD
96	VSS
97	VDD
98	VSS
99	VDD
100	VSS

IF_PMIC

IF PMIC (S2MU004) MAIN CHG

BAT THEM

REPLACE NEAR IF CON

MST

09.22

Battery Connector

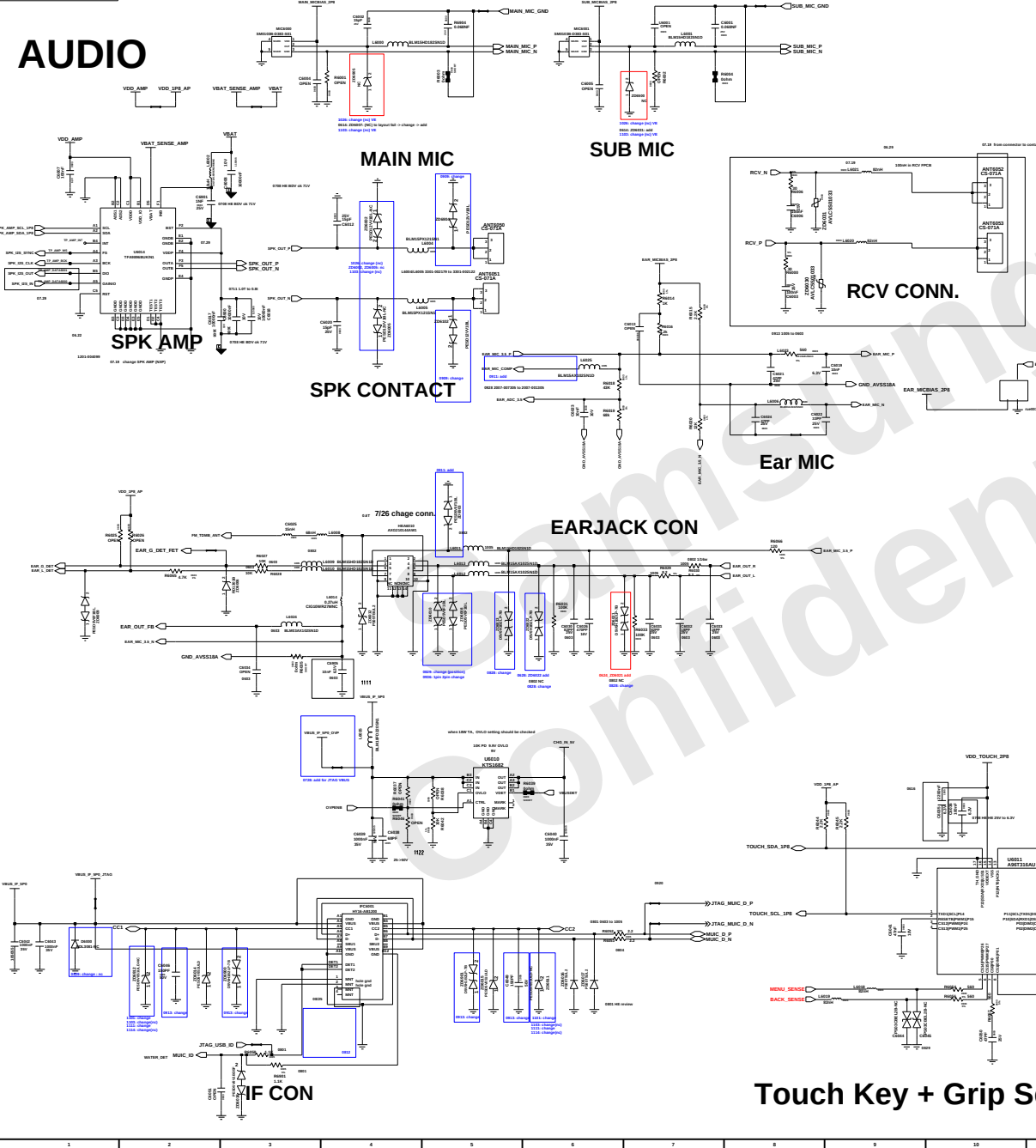
VOLUME KEY

MOTOR

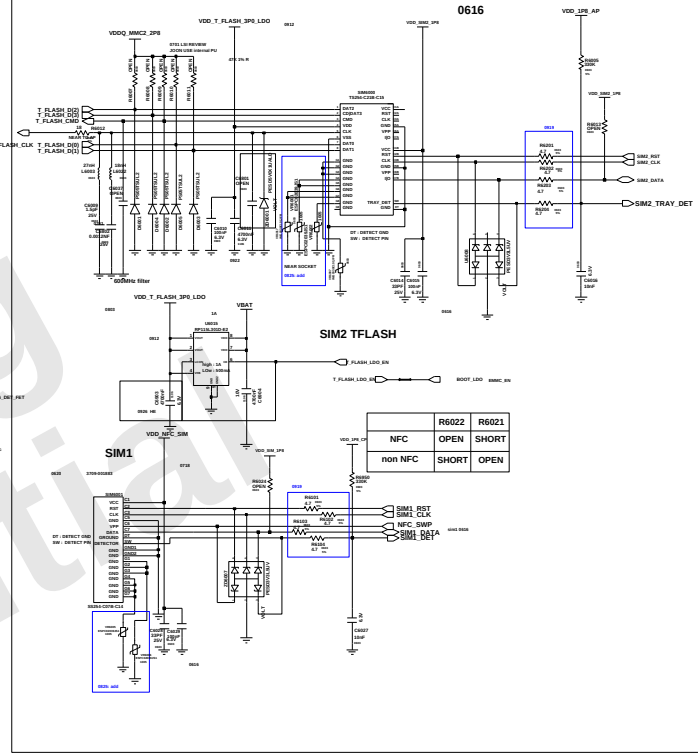
MOTOR LDO

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPERTY INFORMATION THAT IS
UNLAWFULLY DISCLOSED TO A THIRD PARTY.
NO PART OF THIS DOCUMENT IS TO BE
REPRODUCED OR TRANSMITTED IN ANY FORM
OR BY ANY MEANS, ELECTRONIC OR MECHANICAL,
INCLUDING PHOTOCOPYING, RECORDING, OR BY
ANY INFORMATION STORAGE AND RETRIEVAL
SYSTEM, WITHOUT PERMISSION IN WRITING
FROM SAMSUNG ELECTRONICS CO., LTD.

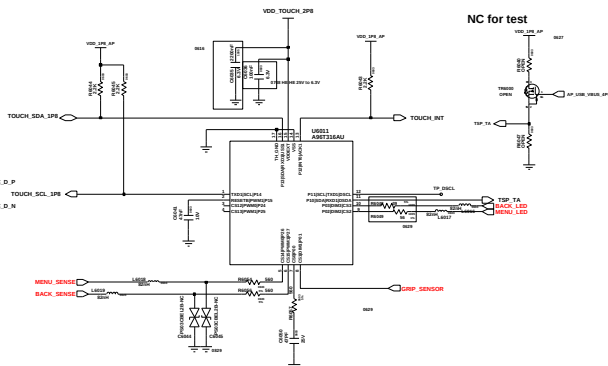
AUDIO



SIM & SD

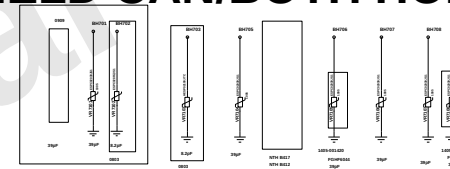


Touch Key + Grip Sensor

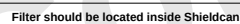


CAMERA

CAMERA



DISPLAY



TDMB for KOR

